PRE-LAB 2 : Function Calls within Functions

> In this pre-lab, we will practice calling functions inside other functions before starting Labwork 2.

LEARNING OBJECTIVES

After completing this pre-lab, students will be able to:

- Define multiple functions with different input types
- Call a function inside another function
- Pass and return values between functions
- Understand how calculation flow moves between functions

EXAMPLE: Nested Function Calls

We will calculate the **area of a rectangle** using 3 functions.

Function 1

Name: `getLength`
Return Type: `double`
Input Parameters: none
Description: Returns the rectangle's length (we assume it's fixed: 10.0)

```
double getLength() {
    return 10.0;
}
```

Function 2

Name: `getWidth`
Return Type: `double`
Input Parameters: none
Description: Returns the rectangle's width (we assume it's fixed: 5.0)

```
double getWidth() {
    return 4.5;
}
```

Function 3

Name: `calculateArea` **Return Type:** `double` **Input Parameters:** none **Description:** Calls the first two functions and returns the area.

```
double calculateArea() {
    double length = getLength();
    double width = getWidth();
    return length * width;
}
```

Main Function

Name: `main` **Return Type:** `int` **Input Parameters:** none **Description:** Calls the third function and prints the area.

```
int main() {
    double area = calculateArea();
    printf("The area of the rectangle is: %lf\n", area);
    return 0;
}
```

Output

Rectangle Area: 45.00